

Jiayi Huang

312-973-9675 | Jiayihuang39@gmail.com | Chicago, IL | <https://github.com/Jiayih4/Projects>

Education

University of Illinois Urbana-Champaign

Bachelor of Science: Computer Science and Economics

Expected May 2025

GPA: 3.6/4.00

Technical Skills

Languages: C, C++, Java, Python, JavaScript, HTML/CSS, SQL, R

Frameworks: React, Node.js, Flask, Django, Bootstrap

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ

Technique: Multithreading, OOP, Database Systems, Machine Learning, Dynamic Programming, Data Analysis

Experiences

iD Tech

May 2024 -

Current

Online Programming Instructor

Remote

- Mentor students aged 13 to 18, in both one-on-one and group lessons of 5-10 students from beginner to advanced programming topics, such as multi-threading, data structures, and algorithms, to create their personalized projects.
- Advised students in collaborative projects, such as developing turn-based battle games, obstacle avoidance games, and conducting data analysis, from scratch to fully functional applications.
- Tests software for bugs and operating speed, fixing bugs and documenting processes to increase efficiency by 18%
- Collaborating with a group of 4 engineers to create project-based lesson plans that teach students Object-Oriented Programming, equipping students with the knowledge for creating their own projects.

The Grainger College of Engineering

Aug 2022 - Jan 2023

Teaching Assistant to An Introduction to Computer Science

Urbana, IL

- Created 12 instructional videos through Zoom, showcasing and explaining the solutions to programming problems, improving 82% of students' grade to 100% for a class of 150 people.
- Assisted 10-15 students with homework and projects in each office hour session with a 100% success rate, improving their grades by 20%
- Reviewed code and conducted testing for lesson plans to eliminate bugs and ensure smooth operations by the students.

Projects

Django Expense Tracker | Python, HTML/CSS, JavaScript, SQLite3, Django, Bootstrap

Jan 2024 - May 2024

Project Lead and Backend Developer

- Developed a full-stack web application using Django framework as the backend and HTML as frontend in a group of 3 engineers.
- Utilized HTML, JavaScript, and CSS with Bootstrap framework to create the front-end webpage with data validation and formatted data entry, preventing the app from crashing due to incorrect data entry.
- Built extensive test coverage, which reduces crash reports by 21%.
- Designed the SQLite3 relational database for user authentication by creating an email and password system that allows each user to have their own secure private expense tracker.

Mosaics | K-D tree, Quick Select, NNS, OOP, C++

Aug 2023 - Jan 2024

Project Lead and Backend Developer

- Implemented a mosaic maker that creates a PhotoMosaic by taking a source picture, dividing it up into rectangular sections, and replacing each section with a small thumbnail image whose color closely approximates the color.
- Designed a K-D tree for storing the LUV color space of each image, improving matching and retrieval of selected image by 50%.
- Utilized quick select to implement a nearest neighbor search algorithm on the K-D tree to quickly retrieve an image most closely matches a source image tile and improving image accuracy by 20%.